

$$\begin{pmatrix} \mathbf{O} & \mathbf{B} \\ \mathbf{C} & \mathbf{O} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{O} & \mathbf{C}^{-1} \\ \mathbf{B}^{-1} & \mathbf{O} \end{pmatrix}$$

and

$$\begin{pmatrix} \mathbf{O} & \mathbf{B} \\ \mathbf{B}^{-1} & \mathbf{O} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{O} & \mathbf{B} \\ \mathbf{B}^{-1} & \mathbf{O} \end{pmatrix}$$

The formula for computing the inverse of a partitioned matrix is as follows:

If $\mathbf{A}$ and $\mathbf{B}$ are non-singular matrices

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{F}^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{F}^{-1} \\ -\mathbf{F}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{F}^{-1} \end{bmatrix} \quad \dots (3.11)$$

where $\mathbf{F} = \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$.

Equivalently, if $\mathbf{D}$ and $\mathbf{G}$ are non-singular,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{G}^{-1} & -\mathbf{G}^{-1}\mathbf{B}\mathbf{D}^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{G}^{-1} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{G}^{-1}\mathbf{B}\mathbf{D}^{-1} \end{bmatrix}$$

where $\mathbf{G} = \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$ provided that the matrix inverses in the expression above are well defined.

3.8 EIGENVALUE AND EIGENVECTOR

Given a square matrix $\mathbf{A}$, if $\mathbf{A}\mathbf{C} = \lambda\mathbf{C}$ for some scalar λ and non-zero vector $\mathbf{C}$, then $\mathbf{C}$ is an eigenvector of $\mathbf{A}$ corresponding to the eigenvalue λ ('Eigen' is a German word that means proper or characteristic). The system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{C} = \mathbf{0}$ has a non-trivial solution if and only if

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0. \quad \dots (3.12)$$

The above equation is called the characteristic equation of $\mathbf{A}$. From this characteristic equation we can solve for the eigenvalues of $\mathbf{A}$. Hence, eigenvalues are also referred to as characteristic roots. Similarly, eigenvectors are called characteristic vectors. You should remember that the eigenvalues and eigenvectors of a real-valued matrix need not be real-valued.

Example 3.6:

Find the characteristic equation and characteristic roots of matrix

$$\mathbf{B} = \begin{pmatrix} 2 & 5 \\ 2 & 3 \end{pmatrix}.$$

For characteristic equation of matrix $\mathbf{B}$ we should have $\det(\mathbf{B} - \lambda\mathbf{I}) = 0$.

Since it is a 2×2 matrix, the $\lambda\mathbf{I}$ matrix will be $\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$

Therefore, the matrix $(\mathbf{B} - \lambda\mathbf{I}) = \begin{pmatrix} 2 & 5 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$

On simplification,

$$\begin{aligned} \det(\mathbf{B} - \lambda\mathbf{I}) &= \begin{vmatrix} 2-\lambda & 5 \\ 2 & 3-\lambda \end{vmatrix} = (2-\lambda)(3-\lambda) - 10 \\ &= 6 - 3\lambda - 2\lambda + \lambda^2 - 10 \end{aligned}$$

By equating it to zero, we obtain $6 - 5\lambda + \lambda^2 - 10 = 0$

By re-arranging terms in the above equation, we obtain the characteristic equation as

$$\lambda^2 - 5\lambda - 4 = 0$$

By solving the above characteristic equation, we obtain the characteristic roots of **B** as follows:

$$\lambda^2 - 4\lambda - \lambda - 4 = 0$$

$$\text{or, } \lambda(\lambda - 4) - 1(\lambda - 4) = 0$$

$$\text{or, } (\lambda - 1)(\lambda - 4) = 0$$

Therefore, the characteristic roots are

$$\lambda = 1 \text{ or } \lambda = 4.$$

When the order of **A** is $n \times n$, the characteristic equation is an n th-order polynomial in λ and has at most n distinct solutions. These solutions (eigenvalues) are usually complex-valued. If some eigenvalues take the same value, there may exist several eigenvectors corresponding to the same eigenvalue. Given an eigenvalue λ , let $C_1, \dots, C_k$ be associated eigenvectors. Then,

$$\mathbf{A}(a_1c_1 + a_2c_2 + \dots + a_kc_k) = \lambda(a_1c_1 + a_2c_2 + \dots + a_kc_k)$$

So that any linear combination of the eigenvectors is an eigenvector corresponding to λ .

That is, these eigenvectors are closed under scalar multiplication and vector addition and form the eigenspace corresponding to λ . As such, for a common eigenvalue, we are mainly concerned with those eigenvectors that are linearly independent.

If **A** ($n \times n$) possesses n distinct eigenvalues, each eigenvalue must correspond to one eigenvector, which is unique up to scalar multiplication. It is therefore typical to normalize eigenvectors such that the Euclidean norm of these vectors is equal to one. It can also be shown that if the eigenvalues of a matrix are all distinct, their associated eigenvectors must be linearly independent. Let **C** denote the matrix of these eigenvectors and **A** denote diagonal matrix with diagonal elements being the eigenvalues of **A**. We can write $\mathbf{AC} = \mathbf{CA}$ as **C** is non-singular. We have

$$\mathbf{C}^{-1}\mathbf{AC} = \mathbf{\Lambda} \text{ or } \mathbf{A} = \mathbf{CAC}^{-1}$$

In this case, **A** is said to be similar to **A** (this symbol is the upper case 'lambda' in Greek alphabet). When **A** has n distinct eigenvalues $\lambda_1 \dots \lambda_n$ (this symbol is the lower case 'lambda') we find that

$$\text{trace}(\mathbf{A}) = \text{trace}(\mathbf{CAC}^{-1}) = \text{trace}(\mathbf{C}^{-1}\mathbf{CA}) = \text{trace}(\mathbf{A})$$

$$\det(\mathbf{A}) = \det(\mathbf{CAC}^{-1}) = \det(\mathbf{A}) \det(\mathbf{C}). \det(\mathbf{C}^{-1}) = \det(\mathbf{A})$$

Let $\mathbf{A}$ be a $n \times n$ matrix with distinct eigenvalues $\lambda_1 \dots \lambda_n$. Then,

$$\det(\mathbf{A}) = \det(\mathbf{\Lambda}) = \prod_{i=1}^n \lambda_i$$

$$\text{trace}(\mathbf{A}) = \text{trace}(\mathbf{\Lambda}) = \sum_{i=1}^n \lambda_i$$

when $\mathbf{A} = \mathbf{C}\mathbf{\Lambda}\mathbf{C}^{-1}$ we have $\mathbf{A}^{-1} = \mathbf{C}\mathbf{\Lambda}^{-1}\mathbf{C}^{-1}$. This shows that the eigenvectors of $\mathbf{A}^{-1}$ are same as those of $\mathbf{A}$, and the corresponding eigenvalues are simply the reciprocals of the eigenvalues of $\mathbf{A}$. similar,

$$\mathbf{A}^2 = (\mathbf{C}\mathbf{\Lambda}\mathbf{C}^{-1})(\mathbf{C}\mathbf{\Lambda}\mathbf{C}^{-1}) = \mathbf{C}\mathbf{\Lambda}^2\mathbf{C}^{-1}$$

The eigenvectors of $\mathbf{A}^2$ are the same as those of $\mathbf{A}$, and the corresponding eigenvalues are the squares of the eigenvalues of $\mathbf{A}$. This result generalizes immediately to $\mathbf{A}^k$.

3.9 CERTAIN SPECIAL MATRICES

Let us discuss about symmetric matrices. Let $\mathbf{C}_1$ and $\mathbf{C}_2$ be two eigenvectors of $\mathbf{A}$ corresponding to the distinct λ_1 and λ_2 respectively. If $\mathbf{A}$ is symmetric, then

$$\mathbf{C}_2' \mathbf{A} \mathbf{C}_1 = \lambda_1 \mathbf{C}_2' \mathbf{C}_1 = \lambda_2 \mathbf{C}_2' \mathbf{C}_1$$

As $\lambda_1 \neq \lambda_2$ it must be true that $\mathbf{C}_2' \mathbf{C}_1 = \mathbf{0}$ so that they are orthogonal. Given linearly independent eigenvectors that correspond to a common eigenvalue, they can also be orthogonalized. Thus, a symmetric matrix is orthogonally diagonalizable, in the sense that

$$\mathbf{C}' \mathbf{A} \mathbf{C} = \mathbf{\Lambda} \text{ or } \mathbf{A} = \mathbf{C} \mathbf{\Lambda} \mathbf{C}'$$

where $\mathbf{\Lambda}$ is again the diagonal matrix of the eigenvalues of $\mathbf{A}$, and $\mathbf{C}$ is the orthogonal matrix of associated eigenvectors.

Let $\mathbf{A}$ ($n \times n$) and $\mathbf{C}$ ($k \times k$) be non-singular matrices. Then for any $n \times k$ matrix $\mathbf{B}$,

$$\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{BC})$$

Thus, the rank of a matrix is preserved under non-singular transformations. Similarly, orthogonal transformation does not change the rank of a matrix. For a symmetric matrix $\mathbf{A}$, $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{\Lambda})$, the number of non-zero eigenvalues of $\mathbf{A}$. Moreover, if we can convert $\mathbf{A}$ to a diagonal matrix, then $\det(\mathbf{A}) = \prod_{i=1}^n \lambda_i$ and $\text{trace}(\mathbf{A}) = \sum_{i=1}^n \lambda_i$ irrespective of the differences in the eigenvalues of $\mathbf{A}$.

Note that the above equality of determinant and trace is valid for a non-singular symmetric matrix, whose eigenvalues are non-zero.

A symmetric matrix $\mathbf{A}$ is said to be positive definite if $\mathbf{b}' \mathbf{A} \mathbf{b} > 0$ for all vectors $\mathbf{b} \neq \mathbf{0}$; $\mathbf{A}$ is said to be positive semi definite if $\mathbf{b}' \mathbf{A} \mathbf{b} \geq 0$ for all $\mathbf{b} \neq \mathbf{0}$. A positive definite matrix thus must be non-singular, but a positive semi-definite matrix may be singular. Suppose $\mathbf{A}$ is a symmetric matrix orthogonally diagonalized as $\mathbf{C}' \mathbf{A} \mathbf{C} = \mathbf{\Lambda}$. If $\mathbf{A}$ is also positive semi-definite, then for any $\mathbf{b} \neq \mathbf{0}$,

$$\mathbf{b}' \mathbf{A} \mathbf{b} = \mathbf{b}' (\mathbf{C}' \mathbf{A} \mathbf{C}) \mathbf{b} = \tilde{\mathbf{b}}' \mathbf{\Lambda} \tilde{\mathbf{b}} \geq 0$$

where $\tilde{\mathbf{b}} = \mathbf{c}\mathbf{b}$. This shows that $\mathbf{A}$ is also positive semi-definite and all the diagonal elements of $\mathbf{A}$ must be non-negative. You can find out that the converse also holds.

A symmetric matrix is positive definite (positive semi-definite), if and only if its eigenvalues are all positive (non-negative). For a symmetric and positive definite $\mathbf{A}$, $\mathbf{A}^{-1/2}$ is such that $(\mathbf{A}^{-1/2})\mathbf{A}^{-1/2} = \mathbf{A}^{-1}$ (Note that $\mathbf{A}^{-1/2}$ is the square root of matrix $\mathbf{A}$). In particular, by orthogonal diagonalization, we obtain

$$\mathbf{A}^{-1} = \mathbf{C}\mathbf{A}^{-1}\mathbf{C}^{-1} = (\mathbf{C}\mathbf{A}^{-1/2}\mathbf{C}')(\mathbf{C}\mathbf{A}^{-1/2}\mathbf{C}') \quad \dots (3.13)$$

So that we may choose $\mathbf{A}^{-1/2} = \mathbf{C}\mathbf{A}^{-1/2}\mathbf{C}'$

The inverse of $\mathbf{A}^{-1/2}$ is $\mathbf{A}^{1/2}$ which is equal to $\mathbf{C}\mathbf{A}^{1/2}\mathbf{C}'$

It follows that $\mathbf{A}^{1/2}(\mathbf{A}^{1/2}) = \mathbf{A}$ and $\mathbf{A}^{-1/2}\mathbf{A}(\mathbf{A}^{-1/2})' = \mathbf{I}$

The square root of a positive semi-definite matrix can be found out by the method of 'Cholesky Decomposition'. Given two symmetric and positive definite matrices $\mathbf{A}$ and $\mathbf{B}$ if $\mathbf{A} - \mathbf{B}$ is positive semi-definite, then so $\mathbf{B}^{-1} - \mathbf{A}^{-1}$.

3.9.1 QR Factorization Matrices

If $\mathbf{A}$ is a real $m \times n$ matrix of rank n , then there exists an $m \times n$ semi-orthogonal matrix $\mathbf{Q}$ (that is, a real matrix satisfying $\mathbf{Q}\mathbf{Q}' = \mathbf{I}_n$) and a $n \times n$ real upper triangular matrix $\mathbf{R}$ with positive diagonal elements. Such that $\mathbf{A} = \mathbf{Q}\mathbf{R}$. The result $q_1, q_2, \dots, q_n$ which are columns of $\mathbf{Q}$ can be obtained by Gram-Schmidt process.

3.9.2 Singular Value Decomposition

Let $\mathbf{A}$ be real $m \times n$ matrix with $\text{rank}(\mathbf{A}) = r > 0$. Then there exist a $m \times r$ matrix $\mathbf{S}$ such that $\mathbf{S}'\mathbf{S} = \mathbf{I}_r$ and $n \times r$ matrix $\mathbf{T}$ such that $\mathbf{T}'\mathbf{T} = \mathbf{I}_r$ and a $r \times r$ diagonal matrix $\mathbf{\Lambda}$ with positive diagonal elements such that $\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{T}'$ since $\mathbf{A}\mathbf{A}'$ is an $m \times m$ symmetric (in fact positive semidefinite) matrix of rank r . The $\mathbf{S}$ and $\mathbf{T}$ can be found by formula $\mathbf{A}\mathbf{A}'\mathbf{S} = \mathbf{S}\mathbf{\Lambda}$ and $\mathbf{T} = \mathbf{A}'\mathbf{S}\mathbf{\Lambda}^{-1/2}$ respectively.

3.9.3 Idempotent Matrix

The matrix $\mathbf{A}$ is said to be idempotent if $\mathbf{A}\mathbf{A} = \mathbf{A}$.

Properties of an idempotent matrix are as follows:

- (i) If $\mathbf{A}$ is idempotent and diagonal, matrix then the diagonal elements 0 and 1 only.
- (ii) If $\mathbf{A}$ is idempotent, $\mathbf{A}', \mathbf{A}^k (k = 1, 2, \dots)$ and $(\mathbf{I} - \mathbf{A})$ are also idempotent.
- (iii) $\mathbf{A}$ and $\mathbf{B}$ are idempotent if and only if $\mathbf{Z} = \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix}$ is idempotent.
- (iv) Eigenvalues of an idempotent matrix are 0 and 1. But the converse is not true.
- (v) For any idempotent matrix $\mathbf{A}$, note that $\text{rank}(\mathbf{A}) = \text{trace}(\mathbf{A})$

Check Your Progress 2

- 1) Show that a non-singular matrix $\mathbf{T}$ exists such that $\mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \mathbf{\Lambda}$ (diagonal) if and only if there is a set of n linearly independent vectors, each of which is an eigenvector of $\mathbf{A}$.

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- 2) Let $\mathbf{X}$ be a matrix of order $n \times k$ and rank k .

Consider the matrix $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

(a) Show that $\mathbf{M}$ is symmetric idempotent.

(b) Show that $\mathbf{MX} = \mathbf{0}$

(c) Show that $\text{rank}(\mathbf{M}) = (n - k)$

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- 3) Let $\mathbf{X}$ be a matrix of order $n \times k$ and rank k , let $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and let $\mathbf{V}$ be positive definite of order n such that $\mathbf{X}'\mathbf{V}^{-1}\mathbf{M} = \mathbf{0}$. Show that

(a) $(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$

(b) $(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$

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3.10 KRONECKER PRODUCT AND VEC-OPERATOR

We usually use the Kronecker product in multiple regression models to write the normal equations in a compact form. The symbol $\otimes$ is used to denote the Kronecker product. Kronecker product is an operation on two matrices that gives us a block matrix.

Suppose we have a $m \times n$ matrix $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ and a $p \times q$ matrix $\mathbf{B}$. The Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is a $mp \times nq$ matrix given by

$$\mathbf{A} \otimes \mathbf{B} = [a_{ij}\mathbf{B}]_{i=1,j=1}^{m,n} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \dots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \dots & a_{2n}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \dots & a_{mn}\mathbf{B} \end{bmatrix} \quad \dots (3.14)$$

Example 3.7:

Find Kronecker product $\mathbf{A} \otimes \mathbf{B}$ for matrices $\mathbf{A}$ and $\mathbf{B}$ given below.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

For the above matrices, the Kronecker product is given by

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 1\mathbf{B} & 2\mathbf{B} \\ 3\mathbf{B} & 4\mathbf{B} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 3 & 0 & 4 \\ 3 & 0 & 4 & 0 \end{pmatrix}$$

Note that the Kronecker product is a mapping such that

$$\mathbb{R}^{m \times n} \times \mathbb{R}^{p \times q} \rightarrow \mathbb{R}^{mp \times nq} \quad \dots (3.15)$$

Further, the Kronecker product of $\mathbf{A}$ and $\mathbf{B}$, i.e., $\mathbf{A} \otimes \mathbf{B}$ is defined for any pair of matrices $\mathbf{A}$ and $\mathbf{B}$ irrespective of their orders. When $n = 1$ and $q = 1$, the Kronecker product of two matrices reduces to the Kronecker product of two column vectors $\mathbf{a} \in \mathbb{R}^m$ and $\mathbf{b} \in \mathbb{R}^p$

$$\mathbf{a} \otimes \mathbf{b} = [a_i \mathbf{b}]_{i=1}^m = \begin{bmatrix} a_1 \mathbf{b} \\ a_m \mathbf{b} \end{bmatrix}$$

The result is a $mp \times 1$ vector. Evidently, the *outer product* of two vectors $\mathbf{x}\mathbf{y} = \mathbf{x}\mathbf{y}^T$ can be also represented by using the Kronecker product as $\mathbf{x}\mathbf{y} = \mathbf{x} \otimes \mathbf{y}^T$.

Example 3.8:

Consider the following vectors and find their Kronecker product.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\text{Hence, } \mathbf{x} \otimes \mathbf{y} = \begin{bmatrix} 1\mathbf{y} \\ 2\mathbf{y} \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

The Kronecker product is also known as the *direct product* or *tensor product*.

Kronecker product has the following properties:

- (i) The Kronecker product of any matrix and a zero matrix is equal to the zero matrix, i.e., $\mathbf{A} \otimes \mathbf{O} = \mathbf{O} \otimes \mathbf{A} = \mathbf{O}$.
- (ii) $\alpha \mathbf{A} \otimes \beta \mathbf{B} = \alpha \beta (\mathbf{A} \otimes \mathbf{B})$, for scalar quantities α and β .
- (iii) $\mathbf{I}_m \otimes \mathbf{I}_n = \mathbf{I}_{mn}$ for $\mathbf{I}_{j \times j}$ is identity matrix of $j \times j$ dimension
- (iv) For matrices $\mathbf{A}_{m \times n}, \mathbf{B}_{n \times k}, \mathbf{C}_{l \times p}, \mathbf{D}_{p \times q}$, we have $(\mathbf{AB}) \otimes (\mathbf{CD}) = (\mathbf{A} \otimes \mathbf{C})(\mathbf{B} \otimes \mathbf{D})$

(v) For matrices $A_{m \times n}, B_{p \times q}, C_{p \times q}$, we have $A \otimes (B \pm C) = A \otimes B \pm A \otimes C$, $(B \pm C) \otimes A = B \otimes A \pm C \otimes A$

(vi) $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$, $(A \otimes B)^T = A^+ \otimes B^+$

(vii) $(A \otimes B)' = A' \otimes B'$

(vii) $\text{rank}(A \otimes B) = \text{rank}(A) \text{rank}(B)$

(viii) $\det(A_{n \times n} \otimes B_{m \times m}) = (\det A)^m (\det B)^n$

(ix) $\text{tr}(A \otimes B) = \text{tr}(A) + \text{tr}(B)$

(x) For matrices $A_{m \times n}, B_{m \times n}, C_{p \times q}, D_{p \times q}$, we have

$$(A + B) \otimes (C + D) = A \otimes C + A \otimes D + B \otimes C + B \otimes D$$

(xi) For matrices $A_{m \times n}, B_{p \times q}, C_{k \times l}$ we have

$$(A \otimes B) \otimes C = A \otimes (B \otimes C)$$

3.10.1 Vec-operator (or Vectorization of Matrix)

The vectorization of a matrix $A \in \mathbb{R}^{m \times n}$ denoted as $\text{vec}(A)$ is a linear transformation which arranges the entries of $A = [a_{ij}]$ as a $mn \times 1$ vector by a column.

$$\text{vec}(A) = [a_{11} \dots a_{m1} \dots a_{1n} \dots a_{mn}]^T$$

A matrix can be arranged as a row vector by stacking the rows. This is known as row-vectorization of a matrix. It is denoted as $\text{rvec}(A)$. It is defined as

$$\text{rvec}(A) = [a_{11} \dots a_{1n} \dots a_{m1} \dots a_{mn}]$$

Example 3.9:

For the matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ find the vec-operator.

$$\text{vec}(A) = [a_{11}, a_{21}, a_{12}, a_{22}]^T \text{ and } \text{rvec}(A) = [a_{11}, a_{12}, a_{21}, a_{22}]$$

The relationship between column vectorization and row vectorization of a matrix can be expressed as follows:

$$\text{rvec}(A) = (\text{vec}(A^T))^T$$

Similarly,

$$\text{vec}(A^T) = \text{rvec}(A)^T$$

For a given $m \times n$ matrix, the two vectors $\text{vec}(A)$ and $\text{vec}(A^T)$ contain the same entries but the order of their entries are different. There is a unique $mn \times mn$ permutation matrix that can transform $\text{vec}(A)$ into $\text{vec}(A^T)$. This permutation matrix is known as the *commutation matrix*. It is denoted by K_{mn} and is defined as $K_{mn} \text{vec}(A) = \text{vec}(A^T)$

Similarly, there is an $nm \times nm$ permutation matrix transforming $vec(A^T)$ into $vec(A)$. Such a commutation matrix denoted K_{nm} , is defined by

$$K_{nm}vec(A^T) = vec(A)$$

Since this formula holds for any $m \times n$ matrix A , we have

$$K_{nm}K_{mn} = I_{mn} \quad \text{or, } K_{mn}^{-1} = K_{nm}$$

3.10.2 Properties of Vec-operator

(i) If A and B are matrices with the same order, then

$$vec(A + B) = vec(A) + vec(B)$$

(ii) $vec(\alpha A) = \alpha vec(A)$, for any scalar quantity α

(iii) $vec(A) = vec(B)$ implies that $A = B$ iff A and B have the same order.

(iv) For any two vectors a and b , $vec(ab)' = b \otimes a$

3.11 MATRIX DIFFERENTIAL

In Unit 7 of MEC 203: Quantitative Methods you were introduced to the concept of vector differentiation. Let us consider two vectors:

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \text{ and } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Let us find out the derivative of $y = a'x$ with respect to x . We find that

$$a'x = a_1x_1 + a_2x_2 + a_3x_3 \quad \dots (3.16)$$

When we take the derivative $\frac{\partial y}{\partial x}$, we actually consider

$$\frac{\partial y}{\partial x} = \begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \frac{\partial y}{\partial x_3} \end{bmatrix}$$

On taking partial derivatives of equation (3.16), we obtain

$$\frac{\partial(a'x)}{\partial x} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = a \quad \dots (3.17)$$

Note that we started with $a'x$, but after differentiation, we obtained a .

Matrix differential is a generalization of a multivariate function differential. The matrix differential (including matrix partial derivative and gradient) has wide applications in optimization theory and econometrics.

Before proceeding further, let us consider the following notations:

$x = [x_1 \dots x_m]^T \in \mathbb{R}^m$ denotes a real vector variable.

$X = [X_1 \dots X_n] \in \mathbb{R}^{m \times n}$ denotes a real matrix variable.

The table below shows classification of real functions.

Function Type	Variable $\mathbf{x} \in \mathbb{R}^m$	Variable $\mathbf{X} \in \mathbb{R}^{m \times n}$
Scalar Function $f \in \mathbb{R}$	$f(\mathbf{x}): \mathbb{R}^m \rightarrow \mathbb{R}$	$f(\mathbf{X}): \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$
Vector Function $f \in \mathbb{R}^p$	$f(\mathbf{x}): \mathbb{R}^m \rightarrow \mathbb{R}^p$	$f(\mathbf{X}): \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^p$
Matrix Function $F \in \mathbb{R}^{p \times q}$	$F(\mathbf{x}): \mathbb{R}^m \rightarrow \mathbb{R}^{p \times q}$	$F(\mathbf{X}): \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$

Let us consider the first case. Here, $f(\mathbf{x}): \mathbb{R}^m \rightarrow \mathbb{R}$ is a real scalar function with a $m \times 1$ real vector $\mathbf{x}$ as variable. The function's output is a scalar: for example, $f(\mathbf{x}) = \text{trace}(\mathbf{x}\mathbf{x}^T)$ where $\mathbf{x}$ is $m \times 1$ vector.

3.11.1 Jacobian Matrix

The Jacobian of a matrix consists of all first order partial derivatives of multivariate function. The row partial derivatives with respect to a $m \times 1$ vector $\mathbf{x}$ are given by

$$D_{\mathbf{x}} = \frac{\partial}{\partial \mathbf{x}^T} = \left[\frac{\partial}{\partial x_1} \cdots \frac{\partial}{\partial x_m} \right]$$

Similarly, the row partial derivatives with respect to a $m \times n$ matrix $\mathbf{X}$ are given by

$$D_{\text{vec} \mathbf{X}} = \frac{\partial}{\partial (\text{vec} \mathbf{X})^T} = \left[\frac{\partial}{\partial x_{11}} \cdots \frac{\partial}{\partial x_{m1}} \cdots \frac{\partial}{\partial x_{1n}} \cdots \frac{\partial}{\partial x_{mn}} \right]$$

When we collect the partial derivatives for all the rows, we obtain the Jacobian matrix, which is given by

$$D_{\mathbf{X}} = \frac{\partial}{\partial \mathbf{X}^T} = \begin{bmatrix} \frac{\partial}{\partial x_{11}} & \cdots & \frac{\partial}{\partial x_{m1}} \\ \vdots & \vdots & \vdots \\ \frac{\partial}{\partial x_{1n}} & \cdots & \frac{\partial}{\partial x_{mn}} \end{bmatrix} \quad \dots (3.18)$$

The partial derivative vector resulting from the operation of $D_{\mathbf{X}}$ on a real scalar function $f(\mathbf{X})$ with $m \times 1$ vector variable $\mathbf{X}$ is an $1 \times m$ row vector and is given by

$$D_{\mathbf{X}} f(\mathbf{X}) = \frac{\partial f(\mathbf{X})}{\partial \mathbf{X}^T} = \left[\frac{\partial f(\mathbf{X})}{\partial x_1} \quad \cdots \quad \frac{\partial f(\mathbf{X})}{\partial x_m} \right]$$

The partial derivative matrix

$$D_{\mathbf{X}} f(\mathbf{X}) = \frac{\partial f(\mathbf{X})}{\partial \mathbf{X}^T} = \begin{bmatrix} \frac{\partial f(\mathbf{X})}{\partial x_{11}} & \cdots & \frac{\partial f(\mathbf{X})}{\partial x_{m1}} \\ \vdots & \vdots & \vdots \\ \frac{\partial f(\mathbf{X})}{\partial x_{1n}} & \cdots & \frac{\partial f(\mathbf{X})}{\partial x_{mn}} \end{bmatrix} \in \mathbb{R}^{n \times m}$$

is known as the Jacobian matrix of the real scalar function $f(X)$, and the partial derivative vector

$$D_{vecX}f(X) = \frac{\partial f(X)}{\partial (vec(X))^T} = \left[\frac{\partial f(X)}{\partial x_{11}} \cdots \frac{\partial f(X)}{\partial x_{m1}} \cdots \frac{\partial f(X)}{\partial x_{1n}} \cdots \frac{\partial f(X)}{\partial x_{mn}} \right]$$

is row partial derivative vector of the real scalar function $f(X)$ with matrix variable X .

Example 3.10:

Determine the Jacobian matrix of the function $f: \mathbb{R}^4 \rightarrow \mathbb{R}$ defined for all $X = (x_1, x_2, x_3, x_4)$ by $f(X) = \|X\|$

$$f(X) = f(x_1, x_2, x_3, x_4) = \|x\|$$

$$= \|(x_1, x_2, x_3, x_4)\| = \sqrt{x_1^2 + x_2^2 + x_3^2 + x_4^2}$$

$$\text{Hence } D_X f(X) = \left[\frac{\partial \|x\|}{\partial x_1}, \frac{\partial \|x\|}{\partial x_2}, \frac{\partial \|x\|}{\partial x_3}, \frac{\partial \|x\|}{\partial x_4} \right]$$

$$= \frac{1}{\sqrt{x_1^2 + x_2^2 + x_3^2 + x_4^2}} [x_1, x_2, x_3, x_4]$$

$$= \frac{1}{\|X\|} (X)$$

3.11.2 Gradient Matrix

Gradient is a term used to define the derivative of a multivariate function. In the case of a single variable, derivative gives us the rate of change in the variable. In the case of multiple variables, there are changes in all the variables. These multiple changes are represented by vectors. As you know, vector has direction. Thus, the gradient matrix will give us the rate of change of the variables.

The gradient vector operator is denoted by the symbol ∇ (read as 'del' or 'nabla'; it is the inverted upper case letter delta). With respect to a $m \times 1$ vector X it is defined as

$$\nabla_X = \left[\frac{\partial}{\partial x_1} \quad \cdots \quad \frac{\partial}{\partial x_m} \right]^T$$

For a $m \times n$ matrix X the gradient operator is defined as

$$\nabla_{vecX} = \frac{\partial}{\partial vecX} = \left[\frac{\partial}{\partial x_{11}} \quad \cdots \quad \frac{\partial}{\partial x_{1n}} \quad \cdots \quad \frac{\partial}{\partial x_{m1}} \quad \cdots \quad \frac{\partial}{\partial x_{mn}} \right]^T$$

The gradient matrix operator with respect to an $m \times n$ matrix X is defined as